

FYP – II PROGRESS REPORT



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1. INTRODUCTION

Government aid distribution programs in Pakistan, such as BISP and Ehsaas, disburse billions of rupees each year to low-income households. Despite their scale, these programs have consistently faced problems, ghost beneficiaries collecting payments they are not entitled to, slow and manual policy enforcement, duplicated household registrations, and limited public visibility into who receives aid and why.

AidLedger was designed to tackle these problems at their root by replacing human-managed eligibility decisions with blockchain enforced smart contract rules. The core idea is that once SBP-inspired policies are encoded on-chain, no single actor can alter them without a traceable consensus event. Beneficiaries are registered through a web portal, their identity attributes are cross-checked against a simulated NADRA dataset and a family-tree table (to prevent duplicate household claims), and the GRC contract produces an immutable eligibility verdict. Approved beneficiaries receive a virtual aid card linked to spending rules that are also enforced on-chain.

FYP-I concentrated on building the foundations: designing and standing up the permissioned Besu network with four institutional nodes (SBP/Governing, BISP/Intermediary, Dummy Bank, Auditor), constructing the synthetic population and KYC dataset, and verifying network stability with the Prometheus-Grafana-Loki-Blockscout monitoring stack. FYP-II picks up from there and targets the complete working prototype. This progress report documents the work completed during the current semester, identifies what remains outstanding, and describes the plan to reach a fully integrated demo.

2. PROJECT TIMELINE

Task	Dur.	Sep	Oct	Nov	Dec	Status
Literature review & requirement analysis	3 wks	✓				Done
SBP regulations study & analysis	2 wks		✓			Done
GRC framework design	4 wks		✓	✓		Done
SRS & SDS documentation	3 wks			✓	✓	Done
Dummy dataset creation (NADRA & Bank)	2 wks				✓	Done
Smart contract development	3 wks				✓	Done

3. PROGRESS

3.1 Milestone 1 Progress (Dec 2025 – Feb 2026)

3.1.1 Smart Contract Development and Deployment

The core GRC smart contracts have been developed in Solidity and deployed to the Besu network via Hardhat. The primary AidPolicy contract stores configurable eligibility parameters (income_threshold, max_household_size, max_monthly_withdrawal, allowed_merchant_categories) and exposes functions for deployPolicy, setPolicyParams, checkEligibility, and validateTransaction. On-chain event logs capture policy activation events and eligibility evaluation outcomes, providing the immutable audit trail required by the GRC model. The BlockchainService integration layer in the backend exposes these contract methods through a clean adapter interface, isolating application code from Web3/Ethereum implementation details.

3.1.2 Backend Application Layer

The Node.js/Express backend exposes REST API endpoints across four domain namespaces: /api/beneficiaries, /api/policies, /api/cards, and /api/transactions. Each namespace is backed by the corresponding application service (RegistrationService, PolicyService, CardService, TransactionService) which orchestrates calls to repository classes and the BlockchainService adapter. JWT-based authentication and RBAC middleware enforce role-level access control at the controller layer. Input validation and error handling are applied uniformly across all endpoints, with blockchain and external adapter calls wrapped in retry logic and descriptive error responses.

3.1.3 Beneficiary Registration and KYC Verification

The registration workflow implements the sequence defined in the SDS: the beneficiary submits CNIC and demographic data through the portal; the RegistrationService validates the CNIC format, matches it against KYC_MASTER to verify identity, checks FAMILY_TREE to detect duplicate family-level aid recipients, and records the result in REGISTRATION with a verified boolean flag. Only verified registrations are forwarded to the GRC policy engine. All KYC verification steps are logged to AUDIT_LOG_DB.

3.1.4 Eligibility Evaluation

Verified registrations are processed by EligibilityService, which calls the on-chain AidPolicy contract via BlockchainService to evaluate the beneficiary against active policy parameters. The smart contract emits an on-chain result (Eligible/NotEligible), which the off-chain validator confirms before EligibilityService writes the outcome to the ELIGIBLE table and triggers the NotificationService to dispatch an Urdu SMS-style message. Bulk re-evaluation is triggered automatically when PolicyService activates a new policy version.

3.2 Milestone 2 Progress (Feb 2026 – Apr 2026)

3.2.1 Card Issuance and POS Transaction Flow

CardService retrieves eligible beneficiaries without issued cards from ELIGIBLE_DB, generates a card token, and persists the card record (card_id, cnic, issue_date, expiry_date) to CARD_ISSUE_DB. The card token serves as the universal identifier whether bound to an NFC UID or a virtual card credential. POS transaction requests include the card_token, merchant_id, merchant_category, and amount; TransactionService validates these parameters against the on-chain GRC contract's spending rules before authorizing or rejecting the transaction, updates the off-chain balance, and logs the outcome immutably. All card and transaction events are logged to AUDIT_LOG_DB and trigger Urdu notifications.

3.2.2 Frontend Portals

Three web interfaces have been developed: the Beneficiary Registration Portal (registration form, eligibility status view, transaction history); the Bank Officer Dashboard (pending application queue, KYC review, card management); and the Admin/Policy Dashboard (policy parameter configuration, bulk re-evaluation trigger, beneficiary and audit overview). All interfaces are built using a lightweight React frontend communicating with the backend via the REST API, with RBAC enforcing role-appropriate views.

3.2.3 Urdu Notification Module

The NotificationService generates Urdu-language notification messages from parameterized templates stored in the database. Notifications are dispatched asynchronously using a mock SMS/email gateway to avoid blocking user-facing transactions. The state machine governing the notification lifecycle — Idle, BuildMessage, Sending, Delivered, Failed, Retry, FailedPermanent — implements a retry strategy with configurable maximum attempts before transitioning to a permanent failure state.

3.3 Milestone 3 Progress (Apr 2026 – May 2026)

3.3.1 FYP-I Validation

Internal validation in FYP-I confirmed: all four Besu nodes successfully join the permissioned network and remain in sync under sustained test-transaction load; Blockscout correctly identifies and attributes transactions from SBP, BISP, bank, and auditor node addresses; Grafana and Prometheus observe stable block times, healthy peer connectivity, and acceptable resource utilization under burst traffic; the synthetic KYC pipeline produces CNICs respecting NADRA length and gender-encoding rules, parent-child family relationships consistent with FAMILY_TREE design, and ML-generated financial attributes within plausible city-cluster distributions.

3.3.2 Smart Contract Testing

Smart contract unit tests were developed using Hardhat's testing framework. Test cases cover: deployment and initialization of AidPolicy contract with default parameters; correct eligibility outcomes for beneficiaries at and around policy thresholds (income, household size, government-job flag, utility bill amount); policy parameter updates and their effect on subsequent checkEligibility calls; transaction validation against spending limits and merchant category restrictions; rejection of invalid inputs (negative thresholds, unknown merchant categories). All deployed contracts were reviewed against common Solidity vulnerability patterns (reentrancy, integer overflow, access control) before deployment to the Besu network.

3.3.3 Backend and Integration Testing

REST API endpoints were tested for correct HTTP status codes, response schemas, and RBAC enforcement across all three roles. Repository interfaces were unit-tested independently of the live PostgreSQL database using mock implementations. End-to-end integration tests exercise the full registration-to-card-issuance pipeline: submitting a registration, verifying KYC, triggering eligibility evaluation against the on-chain contract, confirming the ELIGIBLE table update, and verifying card issuance in CARD_ISSUE_DB. Transaction flow integration tests confirm that approved transactions update off-chain balances and produce immutable on-chain event logs.

3.3.4 Performance Evaluation

Preliminary performance measurements under a simulated concurrent load of 10 users show that registration and eligibility evaluation round-trips complete within the 5-second target specified in the non-functional requirements, accounting for blockchain confirmation latency on the local Besu network. Dashboard read operations served from PostgreSQL return within 500ms. Notification dispatch is fully asynchronous and does not contribute to user-facing latency.

3.3.5 Data Integrity Verification

Cross-checks between on-chain event logs and off-chain AUDIT_LOG_DB records confirm consistency for all policy activation, eligibility evaluation, card issuance, and transaction events exercised during testing. Blockchain-stored policy hashes match the parameter snapshots stored in the off-chain policy table, validating the hybrid on-chain/off-chain integrity model.

4. UPDATED TIMELINE

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5. REFERENCES

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